Supermarket Sales Data Cleaning & Performance Report

Data Pipeline Audit, Lifecycle Standardization & Sales Analysis

DATA SCIENCE AUDIT

INITIAL RECORDS	CLEANED RECORDS	DATA RETENTION	AUDIT TRAIL
8,631	8,514	98.64%	10 Steps
17 Raw Columns	117 Dropped / Imputed	High Quality Output	Fully Standardized

1. LIFECYCLE DATA CLEANING & AUDIT TRAIL

Step 1: Data Ingestion — 8,631 Rows
Loaded raw CSV dataset containing 17 feature columns.

Step 2: Initial Audit — Inconsistency Check
Identified 30 duplicate rows, 55 duplicate Invoice IDs, missing Customer Types (64), Dates (17), and Ratings (88).

Step 3: Exact Duplicates — 8,601 Rows
Removed 30 exact duplicate rows across all feature columns.

Step 4: Invoice ID Resolution — 8,576 Rows
Resolved 25 duplicate Invoice IDs by retaining the latest valid transaction record.

Step 5: Text Standardisation — Clean Fields
Stripped whitespaces and mapped messy Gender and Payment categorical fields to standard Title Case.

Step 6: Quantity Normalisation — Absolute Values
Converted negative transaction quantities into absolute positive values.

Step 7: Date Formatting — Clean Dates
Parsed mixed date formats ('%m/%d/%Y' and '%d-%m-%Y') into standard ISO date column.

Step 8: Revenue Verification — Totals Checked
Verified Total calculations (Unit Price × Quantity × 1.05 tax) and recalculated discrepancies > $0.50.

Step 9: Missing Value Imputation — 8,514 Rows
Imputed Customer Type with 'Unknown', Rating with Median, and dropped 62 rows missing critical Dates.

Step 10: Audit Trail Verification — Verified
Generated final audit log verifying step-by-step row counts from 8,576 down to 8,514.

2. TRADING SHIFTS TRAFFIC ANALYTICS

Morning Shift	Afternoon Shift	Evening Shift
717.33	738.25	852.25
10:00 - 13:00 (Transactions/Hr)	13:00 - 17:00 (Transactions/Hr)	17:00 - 21:00 (Transactions/Hr)

3. QUARTERLY BRANCH SALES TARGET VS ACHIEVEMENT SUMMARY

BRANCH	QUARTER	REVENUE ($)	VARIANCE ($)	% OF TARGET	HIT STATUS
Branch A	2023Q1	$102,248.78	**+$2,248.78**	102.25%	True
Branch A	2023Q2	$90,064.16	**-$9,935.84**	90.06%	False
Branch A	2023Q3	$90,878.34	**-$9,121.66**	90.88%	False
Branch A	2023Q4	$129,821.33	**+$29,821.33**	129.82%	True
Branch A	2024Q1	$112,450.64	**+$12,450.64**	112.45%	True
Branch A	2024Q2	$98,243.28	**-$1,756.72**	98.24%	False
Branch A	2024Q3	$89,175.67	**-$10,824.33**	89.18%	False
Branch A	2024Q4	$117,506.81	**+$17,506.81**	117.51%	True
Branch B	2023Q1	$96,366.01	**-$3,633.99**	96.37%	False
Branch B	2023Q2	$91,993.32	**-$8,006.68**	91.99%	False
Branch B	2023Q3	$84,757.89	**-$15,242.11**	84.76%	False
Branch B	2023Q4	$109,628.69	**+$9,628.69**	109.63%	True
Branch B	2024Q1	$83,503.15	**-$16,496.85**	83.50%	False
Branch B	2024Q2	$79,174.14	**-$20,825.86**	79.17%	False
Branch B	2024Q3	$83,812.84	**-$16,187.16**	83.81%	False
Branch B	2024Q4	$116,106.48	**+$16,106.48**	116.11%	True
Branch C	2023Q1	$100,406.10	**+$406.10**	100.41%	True
Branch C	2023Q2	$82,013.98	**-$17,986.02**	82.01%	False
Branch C	2023Q3	$88,932.03	**-$11,067.97**	88.93%	False
Branch C	2023Q4	$125,174.34	**+$25,174.34**	125.17%	True
Branch C	2024Q1	$104,136.55	**+$4,136.55**	104.14%	True
Branch C	2024Q2	$92,297.57	**-$7,702.43**	92.30%	False
Branch C	2024Q3	$93,115.84	**-$6,884.16**	93.12%	False
Branch C	2024Q4	$140,298.69	**+$40,298.69**	140.30%	True